MMD-MLP: LiDAR-Guided Hyperspectral Data Classification Using Local–Global Directional-MLP With Multiresolution Multiscale Representation

Fulin Luo, *Senior Member, IEEE*, Yiyan Hua, Chuan Fu, Tan Guo, *Member, IEEE*, Yule Duan, and Guangyao Shi

***Abstract*— Hyperspectral images (HSIs) are rich in spectral information and are widely used in the field of land-cover classification. However, existing deep learning methods ignore the combination of multiscale and multiresolution information, while not making better use of light detection and ranging (LiDAR) elevation information to assist in the enhancement of HSI data. To solve the problem above, we proposed a multiresolution, multiscale local–global directional MLP (MMD-MLP) for HSI classification with a LiDAR-guided feature enhancement module in this article. The algorithm first introduced a local–global-directed MLP structure, which effectively combines local and global features. Second, a multiresolution and multiscale feature extraction strategy is invented for the accurate acquisition of the detailed information and features of different land covers under different scale sizes. Subsequently, a LiDAR-guided feature enhancement module, which introduces the elevation information from LiDAR for improving the feature representation of HSI, adopts a cross-attention mechanism to reduce the semantic gap to improve the features of HSI. The proposed algorithm was evaluated on multiple hyperspectral-LiDAR datasets, and the results demonstrate that it achieves state-of-the-art (SOTA) performance. The code will be available at https://github.com/sanxian-svg/MMD-MLP.**

***Index Terms*— Classification, hyperspectral image (HSI), light detection and ranging (LiDAR), multilayer perceptions.**

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I. Introduction

HYPERSPECTRAL image (HSI) provides much more information with hundreds of spectral bands than the traditional RGB image, which has been widely used in many applications [1], [2], [3], including urban construction [4], smart agriculture [5], environmental monitoring [6], and so on [7], [8]. Among these applications, HSI classification is one of the most important and fundamental techniques [9]. However, in HSI, it only contains 1-D spectral and 2-D spatial properties of land covers, which limits the classification performance of land covers with height differences, especially when different land covers share the same spectra information and when the same land cover have different spectral information [10]. For applying the 3-D information, LiDAR data, which focuses on elevation information, have been widely used to effectively guide the classification of HSI [11], [12], [13]. With the additional help of LiDAR data for HSI, higher classification accuracy can be achieved [14], [15], [16]. Therefore, their fusion and interaction of information has become an important research topic [17], [18].

Recent CNN-based research has explored various methods to capture deep features for the classification of HSI and LiDAR data, leading to a series of advances [19], [20]. Xu [19] proposed a spatial–spectral branch framework for HSI, which utilizes the features of LiDAR data to improve the performance of the spatial branch. In [20], researchers introduced a novel network architecture that employs separate CNN branch networks for feature extraction from HSI and LiDAR data. To achieve comprehensive feature representation, Hong [21] designed a joint feature extraction network that integrates encoding and decoding structures with CNNs. Mohla [22] integrated attention mechanisms into CNN architectures, leading to better classification performance. In [23], researchers proposed S2ENet, which uses multiple CNN branches to enhance spectral and spatial features, respectively, improving the final classification performance. He et al. [24] performed spectral angle specialization on the basis of feature extraction and achieved better results by dynamically selecting the window division according to the effect of convolutional feature extraction.

In contrast to CNN-based algorithms whose receptive field is limited in a local region, Transformer has demonstrated

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considerable success in the field of remote sensing [7], [8], [25], particularly due to its global receptive field and long-range relationship capture capabilities [26], [27], [28]. Some researchers have explored a series of Transformer-based classification methods for mixed HSI and LiDAR data [29], [30], [31]. Qing et al. [29] proposed an end-to-end Transformer model for extracting features from hyperspectral and LiDAR data, which has achieved good classification performance. Moreover, researchers introduced a spectral Transformer tailored to extract different features from intra-band groups [30]. Considering the spatial distribution of land covers, in [31], researchers combined traditional morphological theories with a Transformer to improve land-cover classification performance. To fuse features from different levels, Xue et al. [32] proposed a hierarchical Transformer architecture. Zhang et al. [33] utilized a collaboration Transformer with cross-attention of multiple heads for comprehensive feature fusion.

Despite the advantages of Transformer in the field of remote sensing and the progress that has been made, Transformer has the disadvantages and limitations in matrix multiplication with high computational overhead. This facilitates researchers to consider whether it is possible to discard its embedded self-attention mechanism, which is mainly responsible for the high computational cost [34]. This idea has brought the simpler structure of MLP back to the attention of researchers. Unlike the local receptive field of CNNs, MLPs possess a receptive field size equivalent to that of Transformers. However, their computational overhead is lower because of the absence of attention matrix multiplication. Using the low overhead characteristic of MLPs, Tolstikhin et al. [35] introduced a novel backbone network architecture named MLP-Mixer. Expanding on this work, in [36], researchers introduced ResMLP, which integrates residual connections into the original architecture to improve performance. Furthermore, in [37], researchers proposed ViP, which enhances MLP's ability to capture global–local information. Building on the ViP framework, Tang et al. [38] made improvements to Sparse MLP. Due to the powerful representation capabilities of MLPs, some researchers have also introduced MLPs into hyperspectral classification [39], [40], [41]. In [39], researchers introduced an MLP network architecture based on residual learning, called HSI-ResMLP. Meanwhile, Lin et al. [40] combined the U-shaped network structure with MLP, improving the performance of hyperspectral classification. In [41], researchers proposed SSMLP by combining MLP with reciprocal points learning. This algorithm achieved good results in the classification of HSIs.

As another possible improvement to the network backbone, Mamba was proposed in 2023 [41] by integrating a selective scanning mechanism and hardware-aware algorithm into the state-space model (SSM). Mamba performs well in handling long-range information. It can effectively capture global dependencies while ensuring linear complexity. Researchers tried to employ Mamba in the remote sensing field. Hong et al. [42] improved vmamba [43] for the spectral properties of HSIs and proposed HyperMamba. He et al. [44] designed a cross-network based on the interaction of group-wise spectral and spatial features and proposed IGroupSS-Mamba. Based on the quest for higher performance and lower complexity, there is also the exploration of new network backbones such as Kolmogorov–Arnold network (KAN) [45].

Although great progress has been made by the aforementioned works as elaborated, there still exist some problems remaining to be solved. First, existing MLP-based models tend to prioritize spatial information over spectral features and only focus on a local window region, neglecting the global region, which helps enhance feature representation. Second, the same land covers exhibit different properties at different resolutions, which focuses on the detailed textures and edge information of different land covers. At the same time, different land covers have different scales. However, existing MLP models neglect feature information at different resolutions and scales. Finally, it is hard to discriminate between objects with similar spectral characteristics for HSI, which can be solved via auxiliary enhancement from the LiDAR data. However, MLP-based models are currently predominantly confined to HSI data and do not consider how to use LiDAR data to improve the performance of HSI classification.

To solve the above problems, a multiresolution, multiscale local–global directional MLP network called MMD-MLP based on MLP is introduced for HSI classification. MMD-MLP utilizes a new detail representation method, which supplies and enhances the detail information by introducing multiresolution and multiscale features in training. To better extract spectral and spatial information from HSI and achieve harmonization of local and global features, we propose a local–global directional MLP (LGDM) module. Furthermore, a LiDAR-guided feature enhancement module is designed to enhance HSI classification from the guidance of LiDAR data, including LiDAR feature extraction and cross-attention feature enhancement. The major contributions of this article are summarized as follows.

1) To extract spectral and spatial features and integrate the features in local and global regions, we design the LGDM for HSI.
2) To enhance the representation of land-cover details and learn the discriminative features of hyperspectral data, we design a multiresolution multiscale feature extraction strategy.
3) To guide and enhance features using LiDAR data, we develop a LiDAR-guided feature enhancement module based on the cross-attention mechanism to improve HSI classification performance.

The organization of the rest of this article is as follows. The proposed method is described in detail in Section II. Section III conducts some experiments on three HSI-LiDAR datasets. In Section IV, we analyze some conclusions of this article.

II. Methodology

As illustrated in Fig. 1, the proposed MMD-MLP is made up of three principal components: LGDM, multiresolution multiscale feature extraction, and LiDAR-guided feature enhancement. The multiresolution multiscale feature extraction comprises LGDM layers. The LiDAR-guided feature

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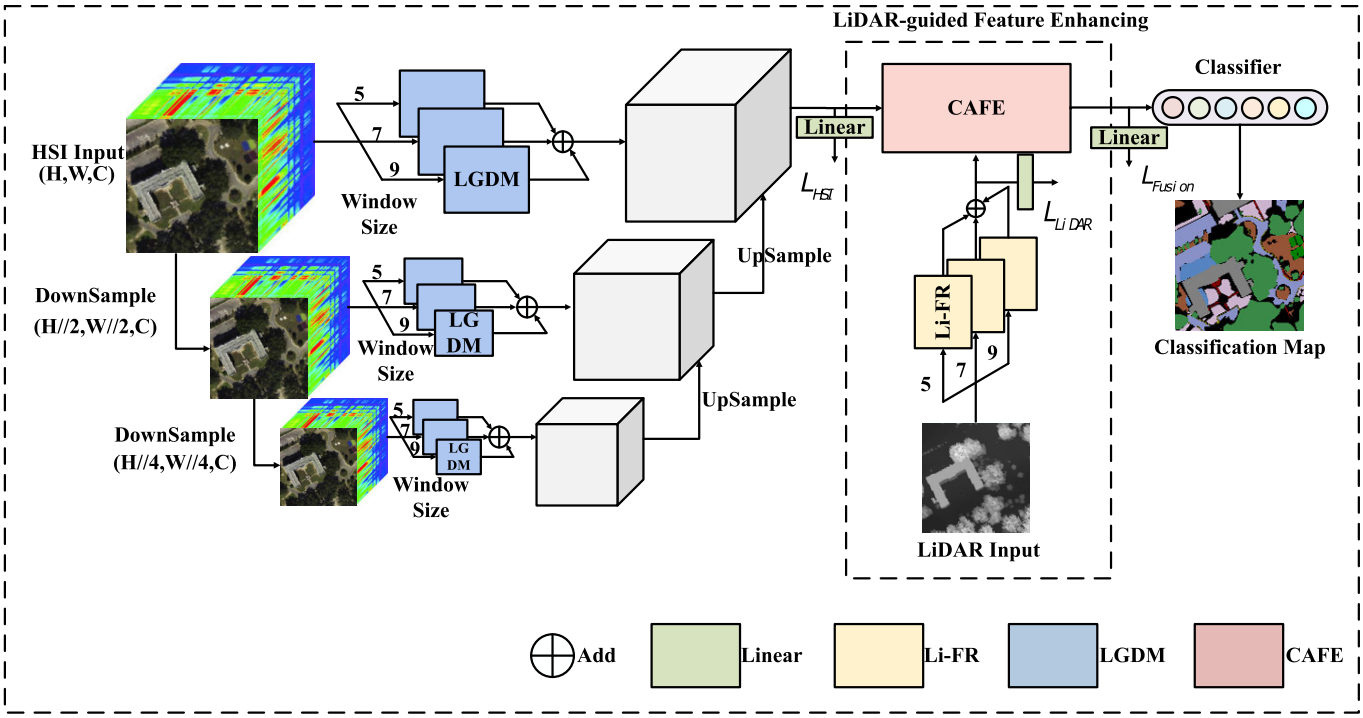


Fig. 1. Structure of the proposed model.

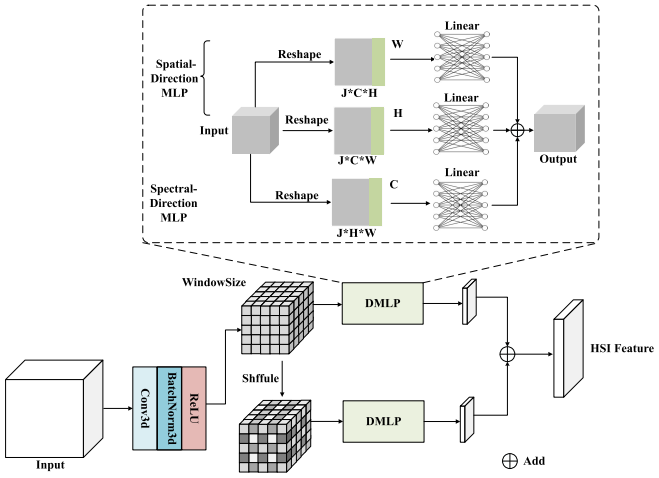


Fig. 2. Structure of LGDM.

enhancement incorporates a LiDAR feature representation (LiFR) module and a cross-attention feature enhancement module (CAFE).

A. Local–Global Directional MLP

Local information can capture the specific characteristics of a given neighborhood, including details such as texture and color. In contrast, global information is more suitable for the extraction of distributions and shapes at the macro-level, facilitating the generation of comprehensive representations. However, the existing MLP-based method generally considers HSI features in a local window region. To fully express the intrinsic properties of HSI, it is essential to consider the combination of local and global information. Thus, we proposed an LGDM module, which combines local and global features in spectral and spatial directions for feature learning. As shown in Fig. 2, 3-D convolution is employed in patch embedding operations to promote the diversity of 3-D features. We denote the initial HSI data as $X^{(H,W,C)}$, where H represents the height, W represents the width, and C represents the channel. The HSI feature tensor X is projected to the high dimension to learn the features in different directions as $X^{(H,W,C,J)}$, where J represents the projection dimension of the features.

Since previous MLP models do not consider the spatial–spectral characteristics of HSI, in LGDM, we first design a direction-wise MLP (DMLP) to learn the feature of HSI, including a spatial DMLP and a spectral DML. Multidirectional MLP learning of hyperspectral data features is performed from vertical, horizontal, and spectral directions, respectively. This is more in line with the data characteristics of hyperspectral data than single-directional MLP learning. Spatial and spectral features are learned more comprehensively and finely, and spatial–spectral interaction capability is improved. Subsequently, the tensor is transformed into $X^{(H*C*J,W)}$ and $X^{(C*W*J,H)}$ through reshaping and expansion, and then the spatial features can be learned by using MLP along the H and W axes, respectively. As illustrated in Fig. 2, the upper section of DMLP depicts the spatial direction of MLP. The lower section of the DMLP illustrates the MLP for the spectral direction. We first compress the high-dimensional feature and spatial features by reshaping the tensor to the shape of $(H*W*J, C)$ using the reshape operation. Then, we flatten the feature along the $H*W*J$ axis, with the aim of learning the spectral features using MLP. Finally, the tensor is reshaped back to its original shape. The whole operation of DMLP can

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be described as follows:

$$\mathbf{DMLP}(X) = \mathbf{MLP}_H(X) + \mathbf{MLP}_W(X) + \mathbf{MLP}_C(X) \quad (1)$$

where $\mathbf{MLP}_H$ includes the MLP operations along the H dimension, $\mathbf{MLP}_W$ includes the MLP operations along the W dimension, and $\mathbf{MLP}_C$ includes the MLP operations along the C dimension, respectively.

The DMLP is capable of extracting both spatial and spectral features with high efficiency. However, its operational range is limited to a local region. In order to learn both local and global regions of HSI in an integrated manner, we propose a local–global extraction strategy. The network is divided into two branches. The first is maintaining the original order, and the second is shuffling the order. The maintained one helps extract the local features. The shuffling inside the input disrupts the order of each patch, which facilitates the synthesis of global information during the training process. In other words, we rearranged the order of patches, which enhances the global interaction capability. Finally, the original data and the shuffled data are fed into the directional MLP for feature extraction, respectively. If we denote HSI data as $X^{(H,W,C,J)}$, the local–global process can be described as follows:

$$X_s = \mathbf{Shuffle}(X) \quad (2)$$

$$\mathbf{LGDM}(X) = [\mathbf{DMLP}(X), \mathbf{DMLP}(X_s)] \quad (3)$$

in which the shuffling operation is described as $\mathbf{Shuffle}(\cdot)$. The DMLP is denoted as $\mathbf{DMLP}(\cdot)$, and [] is the concatenation operator.

B. Multiresolution Multiscale Feature Extraction

Different land covers have different details and scales and exhibit different detail characteristics at different resolutions. To extract diverse detail features and refine edge information, we propose the multiresolution multiscale feature extraction.

In HSI, there are both small objects, such as vehicles, and large objects, such as buildings and roads. To extract features for different object sizes, we propose a multiscale feature extraction approach. As depicted in Fig. 1, we employ three LGDMs to better capture the different scale features of HSI through multiple window sizes. Three LGDMs have window sizes of 5, 7, and 9, respectively, allowing them to focus on land covers with different sizes, resulting in accurate feature representations. After obtaining the feature representations on the three scales, they are summed and combined to obtain the composite features.

We denote **LGDM** operation with a window size of $x * x$ as $\mathbf{LGDM}_{x*x}$, and the formula is shown as follows:

$$\mathbf{MS}(X) = [\mathbf{LGDM}_{5*5}(X), \mathbf{LGDM}_{7*7}(X), \mathbf{LGDM}_{9*9}(X)] \quad (4)$$

where $\mathbf{MS}(\cdot)$ represents the feature learning function for the multiscale operation.

In high resolution, feature extraction focuses on feature details. In low-resolution image, feature extraction focuses on overall shape and edge information. In order to obtain different features at multiple resolutions, we design a multiresolution extraction strategy. In order to facilitate a more comprehensive understanding of the features from the perspective of the overall resolution, the method of whole-image input is adopted, whereby a whole HSI is taken as the input. For HSIs whose dimensions exceed the capacity of the model to be accommodated, the non-overlapping cropping method is employed to divide them. The precise dimensions are delineated in Section III-B. More precisely, each of our epochs is a complete HSI input, and the training/testing is differentiated according to different labeled maps: in the training phase, we compute the loss and perform backpropagation based on the selected training coordinates, and in the testing phase, we statistically record the accuracy of the test coordinates. As shown in Fig. 1, we first perform two down-sampling operations on the hyperspectral data to obtain downsampled images with lower resolutions. Subsequently, multiscale feature extraction is employed to extract features from the original image and the two down-sampled images separately. These features are then merged via concatenation. The features of the down-scaled images need to be up-scaled to the same size as the features from the original image before concatenation. Finally, a skip connection is employed in the concatenation to reduce the information loss caused by the down-sampling. The multiresolution feature extraction process for the HSI images can be represented as follows:

$$X_1 = \mathbf{Downsample}(X), \quad X_2 = \mathbf{Downsample}(X_1) \quad (5)$$

$$\hat{X} = \mathbf{MS}(X), \quad \hat{X_1} = \mathbf{MS}(X_1), \quad \hat{X_2} = \mathbf{MS}(X_2) \quad (6)$$

$$\hat{X_3} = [\mathbf{Upsample}(\hat{X_2}), \hat{X_1}] \quad (7)$$

$$\mathrm{MR}(X) = [\mathbf{Upsample}(\hat{X_3}), \hat{X}, X] \quad (8)$$

where $\mathbf{MS}(\cdot)$ is the multiscale feature extraction process. $\mathbf{MR}(\cdot)$ is the multiresolution process. $\mathbf{Downsample}(\cdot)$ is used to describe convolution down-sampling. $\mathbf{Upsample}(\cdot)$ is the up-sampling of the transposed convolution. [] is the concatenation operator.

C. LiDAR-Guided Feature Enhancement

As mentioned before, the issue of hyperspectral data is that it is challenging to discriminate different land covers with similar spectral characteristics. To address this, we have augmented the spatial feature analysis capability by incorporating LiDAR data. Initially, we use the extracted LiDAR data features and subsequently merge them with HSI features for feature enhancement. This section elaborates on the LiDAR-guided feature enhancement module, which extracts spatial feature information from LiDAR data and fuses HSI and LiDAR information for image classification, including LiFR and CAFE.

1) LiDAR Feature Representation: LiDAR data is represented in the format (H, W, S, K), where S denotes the number of disparate sensors to obtain distance or angle data, and K represents the channel dimension in the patch embedding layer. To learn the LiFRs, we employ 2-D convolution because of the 2-D spatial properties of the LiDAR data. To promote the interaction between features and channels, the LiFR feature extraction module uses the channel dimension

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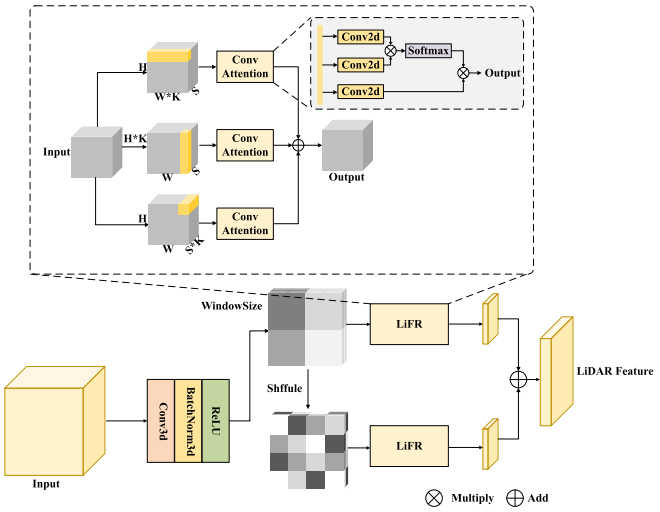


Fig. 3. Structure of LiFR.

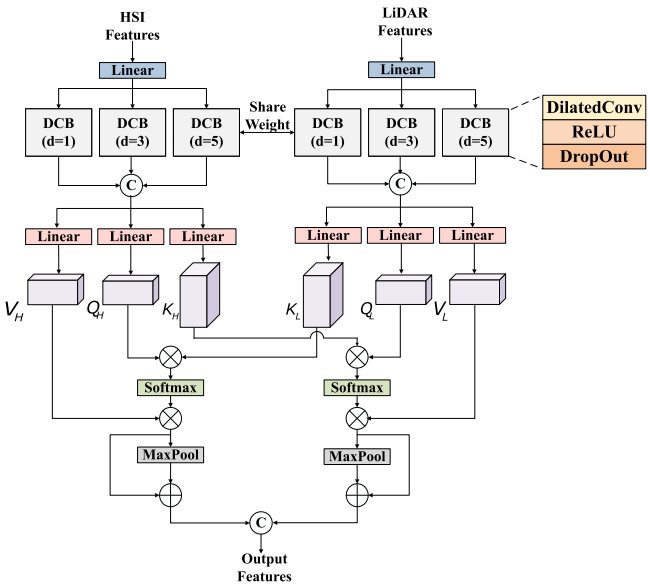


Fig. 4. Structure of CAFE.

K to multiply with the feature dimensions of H, W, and S, respectively, for feature learning with 2-D convolution, which will enhance the interaction and diversity of features. As illustrated in Fig. 3, we first multiply K by H to obtain $L_H^{(W,S,K*H)}$, then we use convolutional attention to extract the features. Similarly, we obtain $L_W^{(H,S,K*W)}$ and $L_C^{(H,W,S*K)}$, learn the features, and then sum them together. After that, LiFR also adopts a combination of local and global features based on shuffle for feature extraction. If we denote LiDAR data as L, the Li-FR module can be described as follows:

$$\mathbf{DCA}(L) = \mathbf{CA}_H(L) + \mathbf{CA}_W(L) + \mathbf{CA}_C(L) \tag{9}$$

$$L_s = \mathbf{Shuffle}(L) \tag{10}$$

$$\mathbf{Li-FR}(L) = [\mathbf{DCA}(L), \mathbf{DCA}(L_s)] \tag{11}$$

where the shuffle operation is described as $\mathbf{Shuffle}(\cdot)$. The convolution-based attention module is denoted as $\mathbf{CA}(\cdot)$, $\mathbf{CA}_{H(\cdot)}$ refers to the channel interaction with the H dimension, $\mathbf{CA}_{W(\cdot)}$ refers to the interaction with the W dimension, and $\mathbf{CA}_{L(\cdot)}$ refers to the interaction with L dimension. We adopt a multiscale feature extraction strategy to synthesize the feature representation at different scales. Here, we denote **LiFR** operation with a window size of $\mathbf{m} * \mathbf{m}$ as $\mathbf{LiFR}_{m*m}$, and the formula is shown as follows:

$$\mathbf{MS}(L) = [\mathbf{LiFR}_{5*5}(L), \mathbf{LiFR}_{7*7}(L), \mathbf{LiFR}_{9*9}(L)] \tag{12}$$

where $\mathbf{LiFR}(\cdot)$ represents the feature extraction. The flow of this module is illustrated in Fig. 3.

Convolutional attention has an advantage over ordinary attention in that it further enhances the local feature by dynamically integrating the global information while preserving the local feature information. Additionally, for LiDAR image data, convolutional attention can more effectively weight local features at different locations to highlight important spatial information. Furthermore, the application of attention does not result in a significant increase in computational overhead due to the relatively small size of the LiDAR dataset. Fig. 3 illustrates the specific structure of multidirectional conv (MD-Conv). The tensor transformation employs a 2-D convolution operation to generate Q, K, and V. These features are then subjected to an attention mechanism for feature learning. We will first introduce the attention mechanism. Multiply L with three learnable variables W_q, W_k, W_v, respectively, to linearly encapsulate L into three matrices: queries Q, keys K, values V. Use Q and K to calculate attention scores while sacling them by the K's dimension and multiply them by V after transforming the score into probability distributions using Softmax. The attention mechanism can be formulated as

$$\mathbf{CA}(L) = \mathbf{Softmax}\left(\frac{QK^T}{\sqrt{d^K}}\right)V. \tag{13}$$

2) CAFE Module: The existence of a semantic gap makes the direct combination of LiDAR and HSI perform poorly. To reduce the semantic gap and use LiDAR features for HSI feature enhancement, we propose the CAFE module, as shown in Fig. 4. To enhance HSI features with LiDAR features, it is necessary to align their distributions and achieve their collaborative representations, so we use two linear layers to project the HSI features X_p and the LiDAR features L_p into the same space, respectively. Then, we introduce multiscale dilation convolutional blocks (MDCBs) for the feature alignment of HSI and LiDAR, sharing the training parameters. MDCB serves to improve the consistency of the features on different scales between HSI and LiDAR data through a unified learning structure. The detailed structure of the MDCB contains a set of dilation convolutions with a dilated rate of {1,3,5}, respectively, which are concatenated and convolved back after obtaining the output results. The objection of dilation convolutions is to improve the size of the receptive field and focus on the interaction of features. The formal operation can be described as follows:

$$\hat{X_p} = [\mathbf{DCB}_{d=1}(X_p), \mathbf{DCB}_{d=3}(X_p), \mathbf{DCB}_{d=5}(X_p)] \tag{14}$$

$$\hat{L_p} = [\mathbf{DCB}_{d=1}(L_p), \mathbf{DCB}_{d=3}(L_p), \mathbf{DCB}_{d=5}(L_p)] \tag{15}$$

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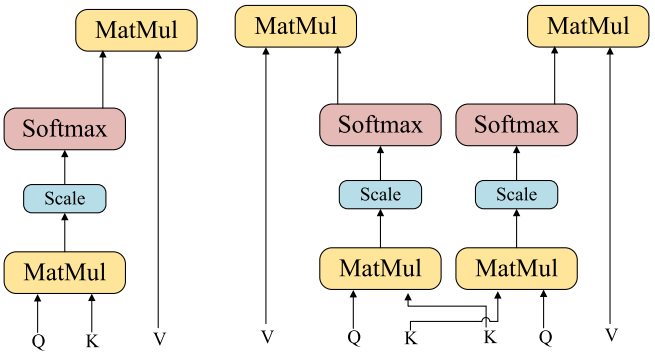


Fig. 5. Attention (left) and cross-attention mechanism (right).

where $\mathrm{DCB}_{d=1}$ stands for dilation convolutional block applying the dilation convolution with a dilation rate of 1.

To facilitate feature interaction between HSI and LiDAR, cross-attention is introduced to promote the LiDAR features to enhance the HSI features. As shown in Fig. 5, the cross-attention mechanism achieves weight interaction of features by exchanging K of the HSI data with K of the LiDAR data. Final_h and final_l are defined by the formula as follows:

$$\mathrm{final}_h = \mathbf{softmax}\left(\frac{Q_h K_l^T}{\sqrt{d^{K_l}}}\right) V_h \tag{16}$$

$$\mathrm{final}_l = \mathbf{softmax}\left(\frac{Q_l K_h^T}{\sqrt{d^{K_h}}}\right) V_l. \tag{17}$$

After obtaining final_h and final_l, we apply 2-D maxpooling to each of them to enhance the significant features. Then, we add them to the input features and concatenate them to obtain $\mathrm{final}_{\mathrm{out}}$. The formula is shown as follows:

$$\mathrm{final}_{\mathrm{out}} = [\mathbf{Maxpool2d}(\mathrm{final}_h) + \mathrm{final}_h, \mathbf{Maxpool2d}(\mathrm{final}_l) + \mathrm{final}_l]. \tag{18}$$

Finally, we project the feature dimension of the output vector into the number of classification categories using the linear layer for classification. The Softmax function is used to obtain the probability of prediction for each individual unit. The predicted label corresponds to the maximum probability value.

D. Loss Function

During model training, the cross-entropy loss function is used in this article. We project the HSI features, LiDAR features, and the fusion features to the same feature dimensions using linear layers, apply the cross-entropy function as a loss function to constrain the three, and adopt the output of the fusion feature as the final classification result. The loss function is combined as follows:

$$\mathrm{Loss} = \mathrm{Loss}_{\mathrm{hsi}} + \mathrm{Loss}_{\mathrm{lidar}} + \mathrm{Loss}_{\mathrm{fusion}} \tag{19}$$

where $\mathrm{Loss}_{\mathrm{hsi}}$, $\mathrm{Loss}_{\mathrm{lidar}}$, and $\mathrm{Loss}_{\mathrm{fusion}}$ denote the cross-entropy for the HSI features, the LiDAR features, and their fusion features, respectively.

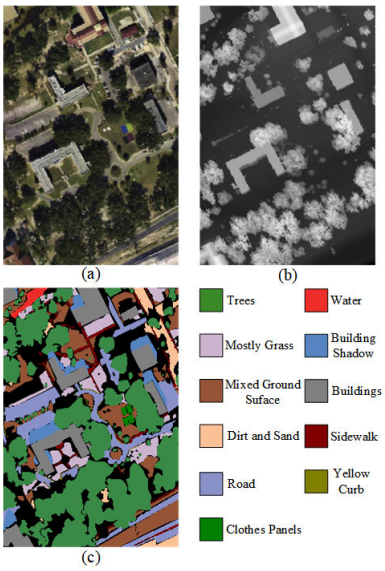


Fig. 6. MUUFL Gulfport. (a) HSI in false color. (b) Grayscale image for LiDAR-based DSM. (c) GT.

TABLE I
NUMBER OF TRAINING AND TEST SAMPLES ON THE MUUFL TABLE

Class	Name	Training	Test
1	Trees	150	23096
2	Mostly grass 2	150	4120
3	Mixed ground surface	150	6732
4	Dirt and sand	150	1676
5	Road	150	6537
6	Water	150	316
7	Building shadow	150	2083
8	Building	150	6090
9	Sidewalk	150	1235
10	Yellow curb	150	33
11	Cloth panels	150	119
Total		1650	52037

III. EXPERIMENTAL RESULTS AND DISCUSSION

In this section, we will verify our method on three hyperspectral and LiDAR datasets (Trento, MUUFL Gulfport, and Houston2018) and compare it with some of the state-of-the-art (SOTA) algorithms.

A. Datasets

1) MUUFL Gulfport: The MUUFL Gulfport dataset was obtained by a reflective optics system imaging spectrometer (ROSIS) sensor at the South Mississippi Harbor Park Campground in 2010, consisting of 72 spectral bands, of which eight spectral bands were discarded due to excessive noise and 64 spectral bands were used for experiments, with a total of 325 × 220 pixels. In addition, the LiDAR data contains two raster data to provide height information. A total of 11 classes are included. The number of train and test samples for each class is shown in Table I. Fig. 6 shows the grayscale map, the pseudo-color map, and the ground truth (GT) of the MUFFL dataset.

2) Trento: The Trento dataset was captured by the AISA Eagle sensor, which included an urban lot in the southern

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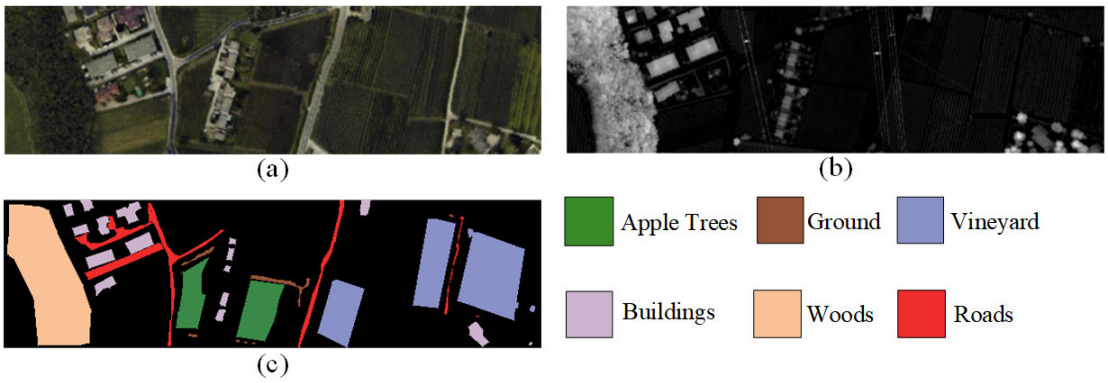


Fig. 7. Trento dataset. (a) HSI in false color. (b) Grayscale image for LiDAR-based DSM. (c) GT.

TABLE II
NUMBER OF TRAINING AND TEST SAMPLES ON THE TRENTO DATASET

Class	Name	Training	Test
1	Apple Trees	129	3905
2	Buildings	125	2778
3	Ground	105	374
4	Woods	154	8969
5	Vineyard	184	10317
6	Roads	122	3052
	Total	819	29395

part of Trento. This HSI data has 63 spectral bands with a wavelength range of 0.43–0.99 μm. The LiDAR data were obtained by an Optech ALTM 3100EA sensor with one raster. The dataset size is 166 $\times$ 600. The spatial resolution is 1 m and contains samples from six different classes. The number of train and test samples for each class is shown in Table II, and Fig. 7 shows the grayscale map, pseudo-color map, and GT of the Trento dataset.

3) Houston2018: The Houston2018 dataset was collected by the HSI Analysis Laboratory and the National Center for Airborne Laser Mapping (NCALM) in Houston. This data is for the 2018 IEEE GRSS Data Fusion Contest. The HSI data include a spectral range of 380–1050 nm, including 48 spectral bands, with a 1-m ground sampling distance, and including 4768 $\times$ 1202 pixels. Fig. 8 shows the grayscale map, the pseudo-color map, and the GT of the Houston2018 dataset. Table III shows the number of train and test samples in each class.

TABLE III
NUMBER OF TRAINING AND TEST SAMPLES ON THE HOUSTON18 DATASET

Class	Name	Training	Test
1	Healthy grass	500	38695
2	Stressed grass	500	129507
3	Artificial turf	68	2667
4	Evergreen trees	500	53821
5	Deciduous trees	500	19671
6	Bare earth	451	17612
7	Water	26	1037
8	Residential buildings	500	158494
9	Non-residential buildings	500	893968
10	Roads	500	182482
11	Sidewalks	500	135234
12	Crosswalks	151	5258
13	Major thoroughfares	500	184937
14	Highways	500	38937
15	Railways	500	27247
16	Paved parking lots	500	45431
17	Unpaved parking lots	14	572
18	Cars	500	25788
19	Trains	500	20978
20	Stadium seats	500	26795
Total		8210	2009131

B. Experimental Setup

A certain number of labeled samples from each dataset are selected as the training set, and the remaining samples are used as the test set.

In this article, we use three evaluation metrics to evaluate the classification results of HSI-LiDAR datasets: overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa).

To validate the effectiveness of the proposed framework, a comparison is made with several representative classification methods, including CNN-based methods such as Coupled-CNN [20], FusAtNet [22], EndNet [21], MDL [42], and Transformer-based methods such as multimodal-transformer (MTNet) [33], Fusion-HCT [43], and S2ENet [23]. The experimental setups of these methods are set according to their source papers. First, we will elaborate on the innovations and features of the compared models. MDL [42] proposes an overarching multimodal deep learning framework. Meanwhile, coupled-CNN [20] is a more classical network that uses two CNNs for HSI and LiDAR feature extraction. FusAtNet [22] exploits HSI and LiDAR features through cross-attention and uses spectral attention and spatial attention fusion for feature representation. EndNet [21], Fusion-HCT [43], and MTNet [33] are three models that use different approaches for the fusion of multimodal information. EndNet uses a deep network structure, Fusion-HCT employs a tandem fusion of CNN and Transformer, and the innovation is the integration of HSI features with LiDAR features at the token level. MTNet uses a spectral Transformer, a spatial Transformer, and a fusion Transformer for HSI and LiDAR feature extraction,

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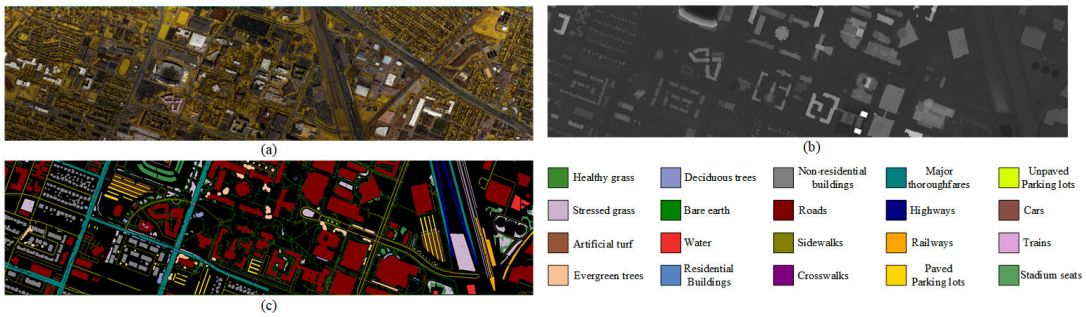


Fig. 8. Houston dataset. (a) HSI in false color. (b) Grayscale image for LiDAR-based DSM. (c) GT.

TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE MUUFL DATASET (%)

Class	Label	CoupledCNN [20]	MDL [42]	EndNet [21]	FusAtNet [22]	S2ENet [23]	MTNet [33]	HCT [43]	Proposed
1	Trees	91.50	85.66	84.97	94.18	90.77	90.41	84.04	**95.47**
2	Mostly Grass	82.03	81.48	80.29	80.70	80.63	84.98	81.26	**88.28**
3	Mixed Ground Surface	78.17	74.47	72.27	83.82	78.28	71.08	64.15	85.75
4	Dirt and Sand	96.36	80.67	91.53	92.89	82.16	**96.36**	92.84	91.35
5	Road	88.35	82.88	89.75	**91.80**	84.50	90.50	75.70	90.05
6	Water	98.73	88.29	**100.0**	99.36	88.29	99.68	**100.0**	99.05
7	Buildings Shadow	94.67	92.99	88.33	90.83	94.82	**95.78**	88.95	93.75
8	Buildings	94.53	88.92	90.02	94.36	88.67	93.69	84.10	**94.63**
9	Sidewalk	83.23	82.19	81.54	80.64	**86.72**	73.85	66.55	84.29
10	Yellow Curb	96.96	**100.0**	96.97	90.91	**100.0**	**100.0**	75.75	93.94
11	Cloth Panels	99.15	**100.0**	98.32	84.87	**100.0**	99.16	100.0	98.31
	AA	89.14	87.05	88.54	89.87	88.62	87.97	83.03	**92.17**
	OA	91.25	84.02	84.54	91.16	87.12	90.50	80.40	**92.05**
	Kappa	85.77	79.44	79.99	88.31	83.23	84.28	74.71	**89.40**

respectively. S2ENet [23] is a spatial–spectral enhancement network that enhances spatial and spectral features.

Before we present the detailed experiments, it is necessary to perform a simple slice of the hyperspectral and LiDAR data. This is because a large data size leads to high computational cost, while a small data size reduces the relevance of global information. Therefore, slicing the HSI and LiDAR data into a 300 × 300 size is more appropriate for the model, which is conducted from experiments. This a compromise way for full-image input on large datasets. The cropping used in this article is a no-overlap cropping rather than the commonly used form of patch overlap. To optimize the network, Adam Optimizer was selected as the initial optimizer. The learning rate was set as 2e–4, and it was halved every 50 epochs for a total of 200 epochs. The experiments with the proposed method and the compared deep learning methods were all implemented on the PyTorch platform and an NVIDIA GeForce RTX 3090Ti GPU with 24 GB of RAM.

C. Classification Comparison and Discussion

Tables III and IV clearly show the OA, AA, Kappa, and per class accuracy of the proposed method and the other comparative methods on the MUUFL Gulfport, the Trento dataset, and the Houston 2018 dataset. The proposed method achieves better results on the MUUFL, Trento, and Houston 2018 datasets, especially on the MUUFL dataset, achieving the more significant breakthrough results.

As shown in Table IV, the proposed model achieves the best OA compared to the other models. Among the CNN-based models compared, CoupledCNN shows the best performance on the MUUFL Gulfport dataset. Compared to FusAtNet and EndNet, CoupledCNN's classification results exhibit more pronounced advantages in Class 4 (dirt and sand) and Class 7 (Buildings Shadow), achieving 96.36% and 94.67%, respectively. This is attributed to the fact that CoupledCNN demonstrates a superior capability to capture the boundary information of features. FusAtNet highlights the importance of the synergy between spectral and spatial features, which is achieved through the combination of spatial and spectral attention. As a result, FusAtNet is more effective in recognizing Class 3 (i.e., Mixed Ground Surface). Among the Transformer-based models, the S2ENet model exhibits the most robust performance, particularly in Class 9 (Sidewalk), with a classification accuracy of 86.72%, which is the highest observed. The proposed model achieves 92.05% OA on the MUUFL Gulfport dataset, better than the models based on CNN and Transformer. The performance improvement over second-best OA is 1.03%, with significant improvement in accuracy on the categories of Class 1 (i.e., trees), Class 2 (i.e., mostly grass), and Class 3 (i.e., ground), achieving 3.97% over CoupledCNN on Class 1 (i.e., trees), 3.3% over MTNet on Class 2 (i.e., mostly grass), and 1.93% over FusAtNet on Class 3 (i.e., ground). This is due to the fact that multiresolution and multiscale feature extraction achieves richer feature representation, resulting in better outcomes.

The classification maps of different models on the MUUFL Gulfport dataset are shown in Fig. 9. The classification map shows that, with the ancillary LiDAR data, the proposed model

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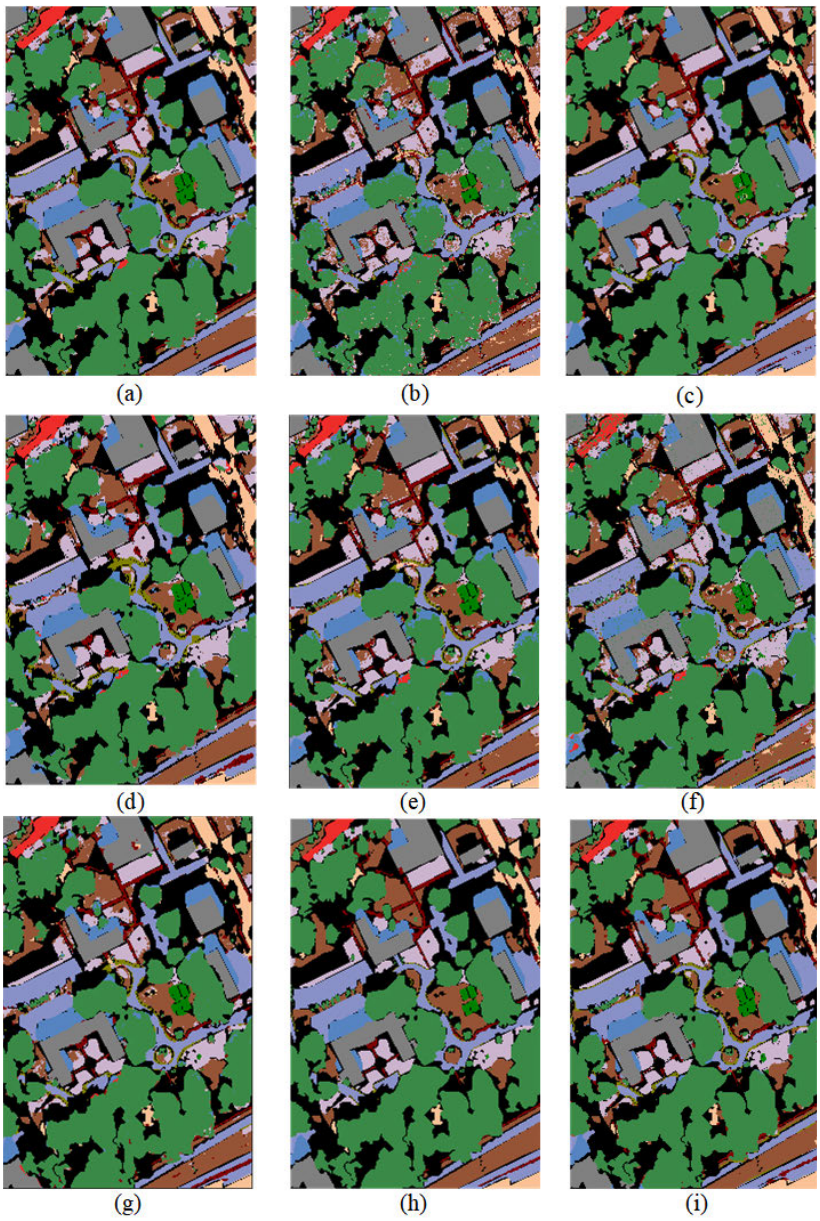


Fig. 9. Classification maps of different methods on the MUUFL dataset. (a) CoupledCNN. (b) EndNet. (c) FusAtNet. (d) Fusion-HCT. (e) MDL. (f) MTNet. (g) S2ENet. (h) GT. (i) Proposed.

TABLE V

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE TRENTO DATASET (%)

Class	Label	CoupledCNN [20]	MDL [42]	EndNet [21]	FusAtNet [22]	S2ENet [23]	MTNet [33]	HCT [43]	Proposed
1	Apple Tree	99.45	99.92	67.89	98.94	**99.99**	98.61	99.56	99.66
2	Building	95.87	95.97	96.16	97.63	98.15	99.06	98.95	**99.12**
3	Ground	100.0	100.0	81.57	**100.0**	99.85	99.70	100.0	**100.0**
4	Wood	**100.0**	99.98	99.83	99.69	100.0	99.92	100.0	**100.0**
5	Vineyard	**100.0**	99.81	94.69	99.88	99.96	99.01	99.93	99.99
6	Roads	98.54	95.80	96.94	96.29	**98.72**	97.09	98.49	98.28
	AA	99.14	98.58	89.51	98.73	99.26	99.05	99.49	**99.51**
	OA	99.48	99.11	91.76	99.12	99.54	98.89	99.63	**99.69**
	Kappa	99.30	98.80	92.78	98.82	99.38	98.72	99.54	**99.58**

achieves excellent resolution at the mixed-ground surface region in the middle of the image and in the mostly grass region at the lower left side of the image, which shows similarity in spectral characteristics.

Compared to other datasets, the Trento dataset has fewer land-cover types and more distinct spectral features, making it easier to achieve higher classification accuracy compared to the other two datasets. As can be seen in Table V, on the Trento

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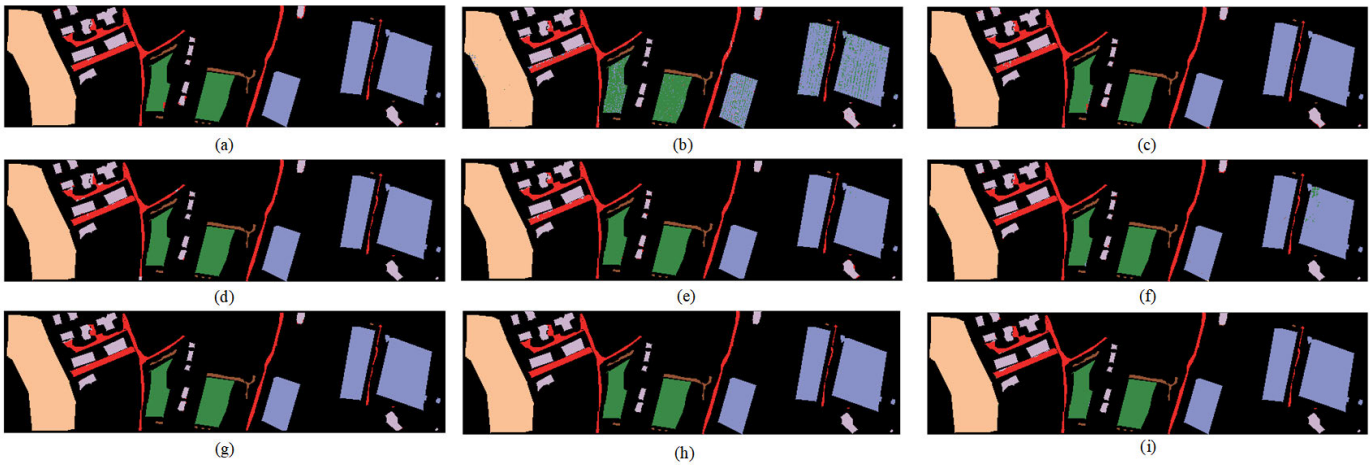


Fig. 10. Classification maps of different methods on the Trento dataset. (a) CoupledCNN. (b) EndNet. (c) FusAtNet. (d) Fusion-HCT. (e) MDL. (f) MTNet. (g) S2ENet. (h) GT. (i) Proposed.

TABLE VI
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE HOUSTON18 DATASET (%)

Class	Label	CoupledCNN [20]	MDL [42]	EndNet [21]	FusAtNet [22]	S2ENet [23]	MTNet [33]	HCT [43]	Proposed
1	Healthy grass	93.39	95.35	94.27	87.11	**96.71**	95.13	94.47	95.31
2	Stressed grass	90.84	92.29	92.31	**93.53**	91.50	92.52	79.46	90.38
3	Artificial turf	98.38	91.67	99.93	**100.0**	91.67	99.63	99.96	97.00
4	Evergreen trees	94.25	97.66	97.36	98.06	**98.57**	98.10	96.07	96.78
5	Deciduous trees	80.62	96.26	96.47	94.45	**97.09**	97.41	95.82	94.98
6	Bare earth	99.48	**99.86**	99.50	99.57	99.76	99.91	98.24	99.64
7	Water	95.83	93.23	95.02	**100.0**	93.70	99.81	100.0	**100.0**
8	Residential buildings	91.43	93.75	91.06	93.98	**95.30**	89.51	88.52	94.22
9	Non-residential buildings	93.93	89.84	86.66	**94.34**	90.68	88.90	88.43	93.78
10	Roads	73.14	68.27	61.24	69.74	72.56	67.98	44.75	**74.56**
11	Sidewalks	78.85	72.00	64.43	71.41	77.06	72.52	55.70	**82.83**
12	Crosswalks	52.38	68.74	61.10	71.10	69.81	58.33	52.20	**87.25**
13	Major thoroughfares	76.08	76.71	72.34	69.32	**77.60**	65.30	51.06	77.38
14	Highways	**98.70**	96.98	94.72	90.96	98.28	96.34	94.14	96.79
15	Railways	99.52	99.60	99.77	85.97	99.70	**99.89**	99.64	99.00
16	Paved parking lots	87.32	97.05	96.37	58.95	**97.39**	96.00	91.46	96.10
17	Unpaved parking lots	**100.0**	**100.0**	99.83	98.80	100.0	99.49	61.43	90.38
18	Cars	89.37	**97.48**	89.67	95.08	**97.48**	94.44	97.31	97.28
19	Trains	76.38	99.17	98.82	**99.76**	99.21	98.39	99.37	98.81
20	Stadium seats	99.83	99.11	99.32	**99.98**	99.10	99.64	99.12	98.88
	AA	88.48	91.25	89.51	92.40	92.16	90.46	84.36	**93.06**
	OA	88.64	87.11	83.74	86.12	88.46	85.26	79.33	**89.58**
	Kappa	85.19	83.62	79.48	82.14	85.32	81.33	73.86	**86.07**

dataset, the performance of Transformer-based models is superior to CNN-based models, because of the better global feature extraction. Among all the comparison algorithms, except for our proposed algorithm, HCT achieved the best classification performance, indicating that the token-level integration in HCT is a superior method of information interaction compared to the separated spatial and spectral Transformers. In addition to HCT, S2ENet also performed well. This algorithm achieves cross-modal fusion by scaling the size of HSI and LiDAR to unify the images of both modalities in the same dimension, thus showing superior OA classification accuracy. Compared to the other algorithms, our algorithm achieved better classification performance. Similar conclusions can also be drawn from the classification map of these algorithms on the Trento dataset, as shown in Fig. 10. For the classification maps of the other comparison algorithms, there is some confusion between the boundaries of buildings and roads, while our proposed model achieved a clearer distinction between them.

The experimental results on the Houston18 dataset further demonstrate the effectiveness of our algorithm. As shown in Table VI, for the challenging Houston18 dataset, the proposed model still achieves an OA of 89.69%, outperforming all the comparison algorithms. Thanks to multiscale feature extraction, the proposed model is still able to effectively extract detailed features from large-scale datasets. Among the comparison algorithms, S2ENet performs well in the classification of certain categories, indicating that its spatial–spectral cross-modality enhancement module plays an important role in improving OA. Due to the large scale of this dataset, there are many different scales of ground objects. Thanks to the ability of multiresolution to extract information about ground objects of different sizes more effectively, the proposed algorithm can

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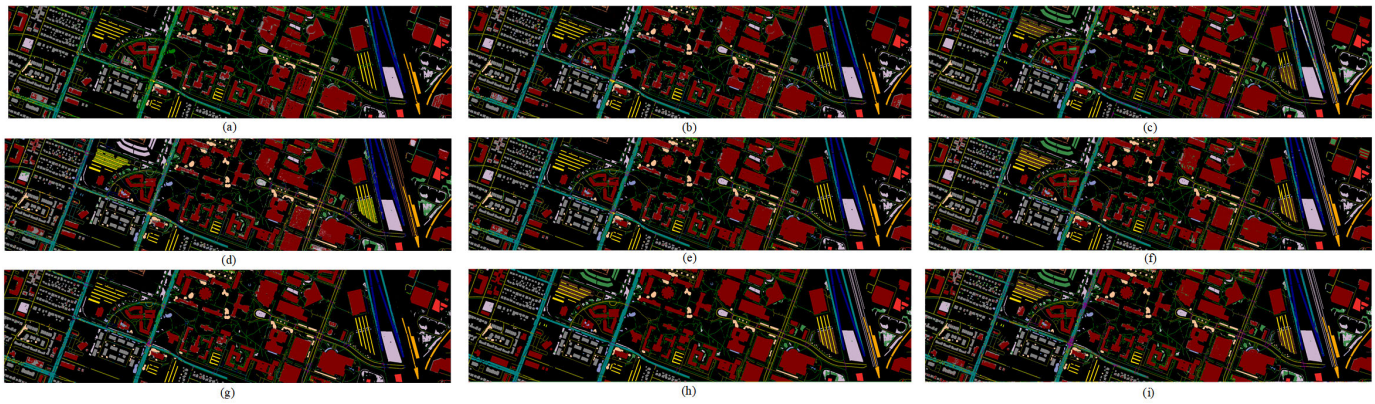


Fig. 11. Classification maps of different methods on the HOUSTON dataset. (a) CoupledCNN. (b) EndNet. (c) FusAtNet. (d) Fusion-HCT. (e) MDL. (f) MTNet. (g) S2ENet. (h) GT. (i) Proposed.

TABLE VII
RESULTS OF CLASSIFICATION IN ABLATION STUDIES OF THE NETWORK WE PROPOSED

	Metrics	Baseline	A	B	C	D	E	F	G	H	I	J
Local-Global			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Multi-Resolution				✓			✓	✓	✓	✓	✓	✓
Multi-Scale					✓		✓	✓	✓	✓	✓	✓
Directional MLP						✓	✓	✓	✓	✓	✓	✓
LiFR	simple							✓	✓	✓	✓	✓
	Shuffle								✓		✓	✓
	multi-scale									✓	✓	✓
CAFE												✓
MUUFL	AA(%)	76.72	77.14	80.73	80.96	81.41	82.24	83.38	85.17	85.62	86.69	**92.17**
	OA(%)	75.62	76.56	81.90	80.62	79.79	82.18	82.37	82.84	85.88	86.97	**92.05**
	Kappa(%)	68.62	69.79	76.39	74.92	73.89	76.86	80.09	80.47	80.70	81.42	**89.40**
Trento	AA(%)	91.48	91.72	98.33	95.47	94.05	97.89	98.78	98.85	98.95	98.99	**99.46**
	OA(%)	93.77	93.86	98.63	96.53	95.35	98.39	99.23	99.30	99.36	99.37	**99.69**
	Kappa(%)	91.67	91.78	98.16	95.36	93.78	97.84	98.97	99.06	99.14	99.15	**99.57**

TABLE VIII
MODEL PARAMS AND COMPUTATIONAL OVERHEAD OF ALL METHODS ON MUUFL GULFPORT

Model	CoupledCNN	MDL	EndNet	S2ENet	HCT	MTNet	FusAtNet	DHViT	Proposed
Params	113.15K	104.55K	86.72K	173.14K	1427K	296.90K	35398.38K	5140.48K	7851.43K
Flops	3.68M	1.70M	87.09K	8.29M	2.62M	23.96M	3227M	88.76M	66.92M

acquire edge information better, and the boundaries are clearer. The classification map on the Houston18 dataset is shown in Fig. 11. And the edge details of the proposed algorithm are clearer and there are fewer classification errors.

D. Ablation Study

To verify the effectiveness of the proposed modules, we use two datasets to analyze the ablation performance of the proposed network. The ablation experiments includes six parts: local–global interaction, multiresolution, multiscale, directional MLP, LiFR, and CAFE modules. The baseline is specified to be a single-direction feature learning under a one-scale window with one-single resolution without shuffling. Table VII shows the performance of each component. We incrementally add each part to the baseline to evaluate its performance on the Trento and MUUFL Gulfport datasets. Among these, LiFR is composed of three parts: a simple module, a shuffle operation, and a multiscale operation, which can also be evaluated in the ablation experiments.

As shown in Table VII, the local–global module improves slightly on the Trento dataset, while it improves more significantly on the MUUFL Gulfport, due to the fact that the Trento dataset is simple to some extent. The performance of the directional MLP feature learning alone optimizes local–global module by 3.23%, the multiscale window alone increases the OA by 4.06%, and then the addition of multiresolution information alone increases the OA by 5.34%. It is evident that multiresolution and multiscale feature extraction plays a significant role in the training process. Simple LiFR shows small improvements, while the shuffle operation and the multiscale operation show performance improvements, respectively. The

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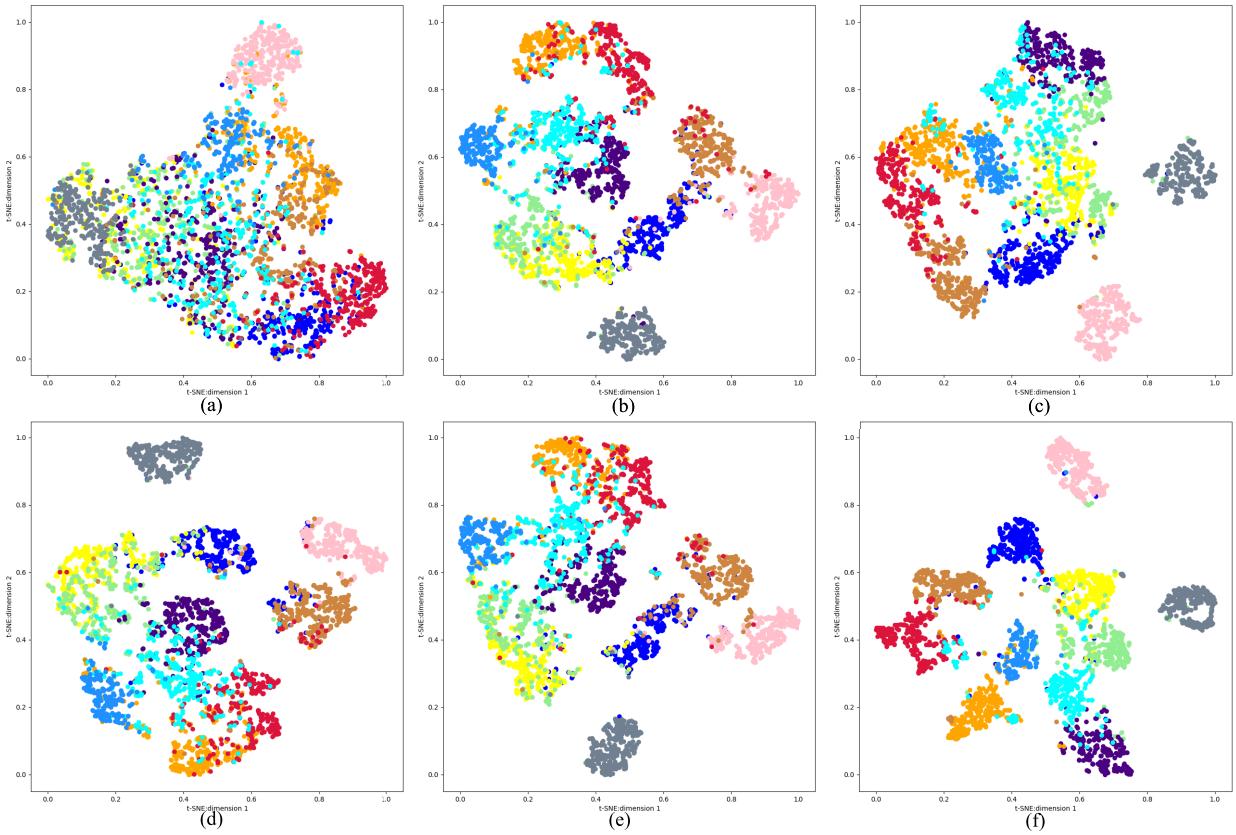


Fig. 12. t-SNE scatterplots of different conditions on MUUFL dataset. (a) Only LiDAR data. (b) Only HSI data. (c) LiDAR-guided HSI without CAFE. (d) Without directional MLP. (e) Without multiresolution and multiscale. (f) LiDAR-guided HSI with CAFE.

accuracy under Simple LiFR is not significantly improved, but with the addition of the refined extraction modules, the performance increases a lot. This demonstrates that LiFR and CAFE are effective in feature extraction and reduction of the semantic gap. In the ablation experiment on the MUUFL Gulfport, the model achieves only 82.18% OA with HSI data alone. However, after introducing LiDAR data, the accuracy of the model increases to 86.97%. This indicates that HSI data alone are not sufficient for the model on the MUUFL Gulfport, and the LiDAR data is necessary as complementary data. The model is more effective when LiDAR data is added. The addition of the CAFE module results in a more significant effect, increasing the accuracy from 86.97% to 92.05%. Since the Trento data itself is relatively simple, simply combining the HSI data with the LiDAR data yields a 99.23% OA. Therefore, the CAFE module performs better on the MUUFL Gulfport dataset but not as significantly on the Trento dataset. The ablation analysis further demonstrates the necessity of each module in the model.

In order to more effectively represent the influence of the LiDAR data and each module on the final classification results, we generated t-SNE scatterplots. The results are shown in Fig. 12. Fig. 12(a)–(d) illustrates the scatterplots generated for pure HSI data, pure LiDAR data, HSI-LiDAR without fusion module, and HSI-LiDAR with fusion module, respectively. Fig. 12(e) and (f) illustrates the scatterplot with the multiresolution multiscale module and the multidirectional MLP feature extraction, respectively. From the t-SNE scatterplots, we can find that our module plays an important role for better feature extraction.

E. Model Params and Complexity

In order to analyze the computational complexity of the different models, we calculate the parameters and FLOPs of the proposed model and the comparison model. It can be seen from Table VIII that our model has a higher number of parameters and FLOPs than CNN-based models and some Transformer-based models. This is due to the fact that we take the whole image as input. The operation of a large matrix leads to the growth of computational overheads. However, due to the low complexity of MLP, the computational overhead of our model is still smaller than that of some Transformer-based models. Moreover, we stress that the model was trained and evaluated on specific hardware configurations and that the calculation requirements may vary on other hardware.

IV. Conclusion

In this article, we propose a LiDAR-guided LGDM model based on multiresolution and multiscale for HSI classification. In this algorithm, we extract diverse detail features and refine edge information by adopting multiresolution and multiscale strategies. We have also innovatively designed an LGDM to better integrate local and global features. In addition, we have

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constructed a LiDAR-guided feature enhancement module through a cross-attention mechanism to reduce the semantic gap between hyperspectral data and LiDAR data. Experiments on the MUUFL, Trento, and Houston2018 datasets have verified the effectiveness of our model.

Although the proposed algorithm has achieved good classification accuracy, the complexity of the model can still be further optimized. In addition, our algorithm has not fully utilized a large number of unlabeled samples. Making full use of the information from these unlabeled samples can alleviate the problem of insufficient quantity of labeled samples to some extent. In the future, we will focus on how to utilize these unlabeled samples to improve the final classification accuracy.

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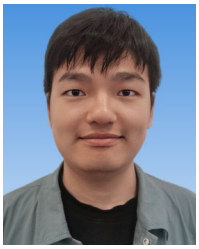
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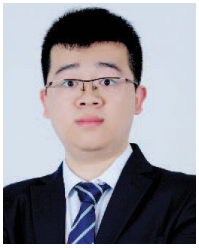
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